Recent Technology in 3D Visualisation: Representations, Converters and Camera Models

Darin Johnson · October 2026 · one-page report accompanying the from-scratch 3D viewer (camera module, object loader, mouse manipulation)

1. 3D representation

Explicit representations are still the workhorses of every pipeline: polygon meshes (OBJ, STL, PLY, glTF), point clouds from LiDAR and photogrammetry, and voxel grids. What changed in the last five years is the arrival of *learned, differentiable* representations. Neural Radiance Fields (NeRF, Mildenhall et al., 2020) store a scene in an MLP that maps a 3D position and viewing direction to colour and density, rendered by volumetric ray marching; Instant-NGP (Müller et al., 2022) cut training from hours to seconds with multi-resolution hash grids. 3D Gaussian Splatting (3DGS, Kerbl et al., 2023) replaced the implicit field with millions of explicit anisotropic Gaussians that are rasterised ("splatted") in real time, and it has largely displaced NeRF for novel-view synthesis; surveys now track hundreds of variants (2DGS for accurate surfaces, 4D Gaussians for dynamic scenes, compressed and streamable splats). Signed-distance and occupancy fields (DeepSDF, NeuS) remain the standard when a watertight surface is required.

In 2024 to 2026 the frontier moved from *reconstruction* to *feed-forward generation*. TRELLIS (Microsoft, 2024) introduced a structured latent that decodes into a mesh, a radiance field or Gaussians from one network; TRELLIS.2 (4B parameters, 2025) and Hunyuan3D 2.x (Tencent, 2025) turn a single image into a PBR-textured mesh in seconds. Apple's SHARP (ICLR 2026) regresses a metric Gaussian scene from one photograph in under a second, and "world models" such as World Labs' Marble (public since Nov 2025, v1.1 Apr 2026) generate whole navigable environments and export them as Gaussian splats or meshes. The practical consequence for a viewer is that triangles are no longer the only primitive: point and Gaussian primitives must be first-class citizens.

2. Converters

Interchange still runs through a handful of formats. OBJ (text, ubiquitous, no PBR materials), STL (loose triangles, 3D printing), PLY (typed property lists; the de-facto container for point clouds and, lately, raw Gaussian splats), glTF 2.0 (Khronos, 2017; the "JPEG of 3D" for web and AR, with PBR materials and Draco/meshopt compression), FBX (proprietary) and OpenUSD (Pixar). Two standardisation milestones define the current moment. The Alliance for OpenUSD ratified the **OpenUSD Core Specification 1.0** in December 2025, the first formal definition of USD's composition engine, cementing USD as the interchange backbone for film, NVIDIA Omniverse and Apple visionOS (USDZ). Khronos published **KHR_gaussian_splatting** for glTF as a release candidate in early 2026 together with an SPZ compression extension; Niantic's open-source `.spz` container is roughly ten times smaller than an equivalent `.ply`, and CesiumJS, ArcGIS and Scaniverse already read it.

On the tooling side, Assimp (40+ formats), Blender's scriptable importers and exporters, Python's trimesh, Open3D and meshio, glTF-Transform and gltfpack for web optimisation, and the `usdcat`/`usdzip` tools cover file-level conversion. The harder and more active problem is conversion *between representations*: marching cubes and Poisson reconstruction (points or fields to mesh), SuGaR and 2DGS (Gaussian splats to mesh), baking NeRFs into meshes for real-time use, and mesh-to-SDF for simulation and collision. The `convert.py` script in this repository implements the simplest tier (OBJ, STL and PLY in both directions) on top of the viewer's own loader.

3. Camera models

The pinhole model, $\mathbf{x} = K\,[R \mid t]\,\mathbf{X}$ with $K = [[f_x, 0, c_x], [0, f_y, c_y], [0, 0, 1]]$, is still the foundation. The viewer's camera module builds exactly this: `look_at` produces $[R \mid t]$ (the extrinsics) and the OpenGL projection matrix is a re-parameterisation of K with near and far planes (`perspective_from_intrinsics`). Real lenses add distortion models on top: Brown-Conrady radial/tangential (OpenCV), Kannala-Brandt and other fisheye models for wide fields of view, equirectangular projection for 360° cameras, and the rolling-shutter and generic per-pixel-ray models used by COLMAP and SLAM systems. Phone camera modules now ship with LiDAR or time-of-flight depth, and event cameras require asynchronous models of their own.

Three research trends are reshaping how cameras are handled. First, cameras became *learnable*: BARF (2021) refines poses jointly with a NeRF, and most 3DGS pipelines optimise camera parameters during training. Second, *ray-based parameterisations*: "Cameras as Rays" (Zhang et al., ICLR 2024) represents a camera as a bundle of Plücker rays predicted by a diffusion model instead of a rotation and translation. Third, *pose-free, feed-forward reconstruction*: DUSt3R (CVPR 2024) and MASt3R (ECCV 2024) regress point maps from uncalibrated image pairs; VGGT (Wang et al., CVPR 2025 Best Paper) predicts intrinsics, extrinsics, depth and point tracks for hundreds of views in a single forward pass in under a second; MapAnything (Meta/CMU, 3DV 2026) unifies calibrated and uncalibrated SfM, multi-view stereo and depth completion with optional intrinsics or poses as inputs. Calibration is turning from a prerequisite into an optional hint, yet the K-matrix abstraction remains the interface every tool speaks, which is why the viewer exposes `intrinsics()` and `extrinsics()`.

References

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